

The background features a stylized American flag. Red and white stripes curve from the top left towards the center. A blue field with white stars is in the top right corner. A row of white stars runs along the bottom edge.

PRISON GERRYMANDERING IN TENNESSEE

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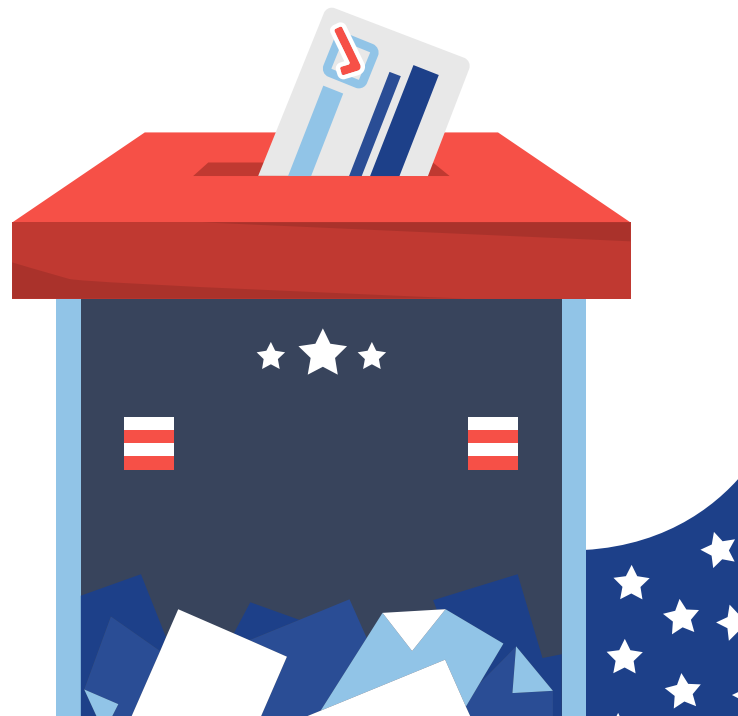
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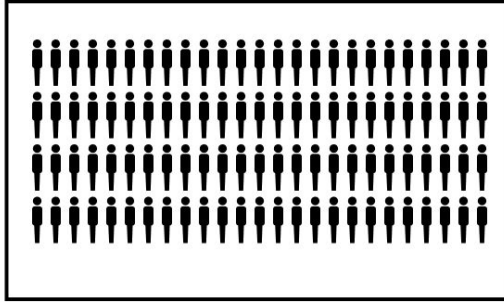
WHAT IS PRISON GERRYMANDERING?

- Also known as prison malapportionment
- “[A] practice of distorting representation by counting incarcerated people in the jail or prison where they are confined, rather than at their place of residence”^[1]

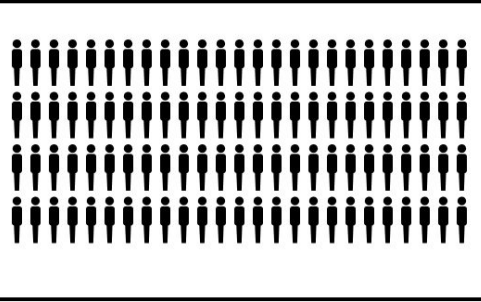


PRISON GERRYMANDERING EXAMPLE^[1]

A

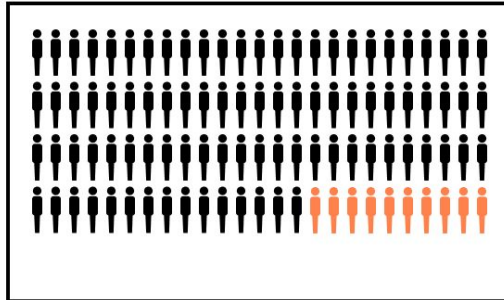


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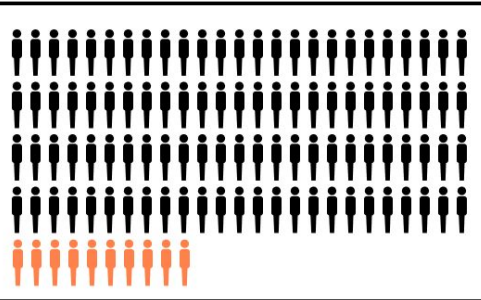


District A and
District B each
have 100
residents

A

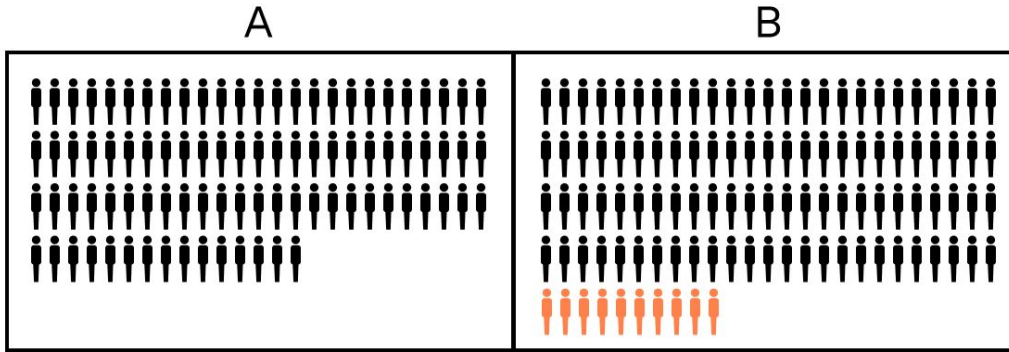


B

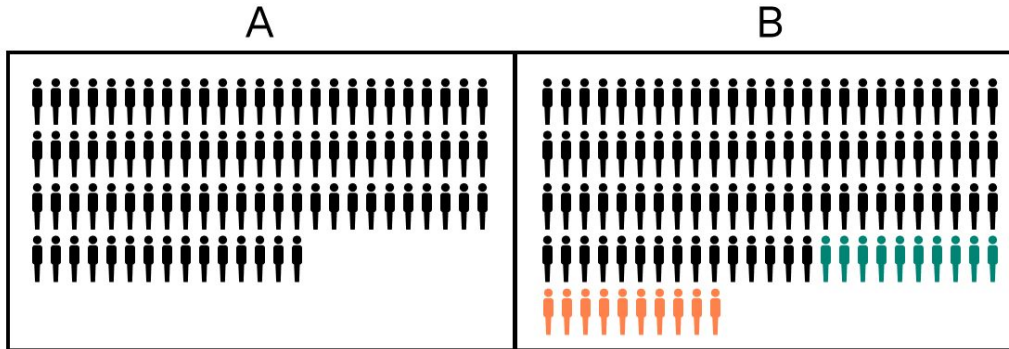


10 individuals
from district A
get incarcerated
in district B

PRISON GERRYMANDERING EXAMPLE^[1]

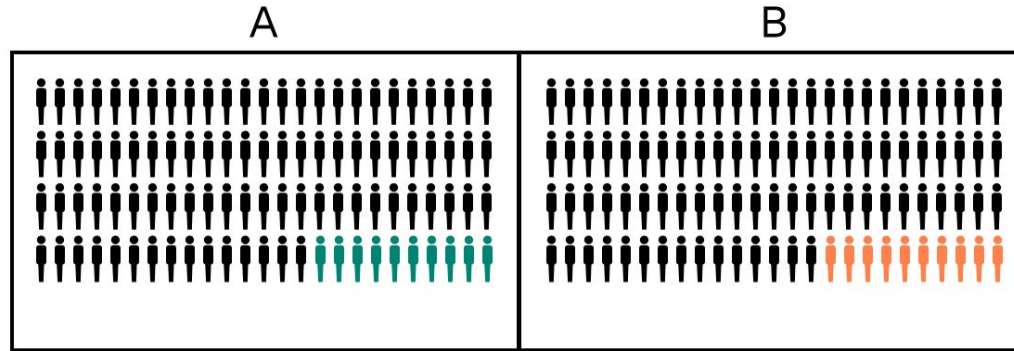


The 10 individuals are moved to district B because that is where the prison is located

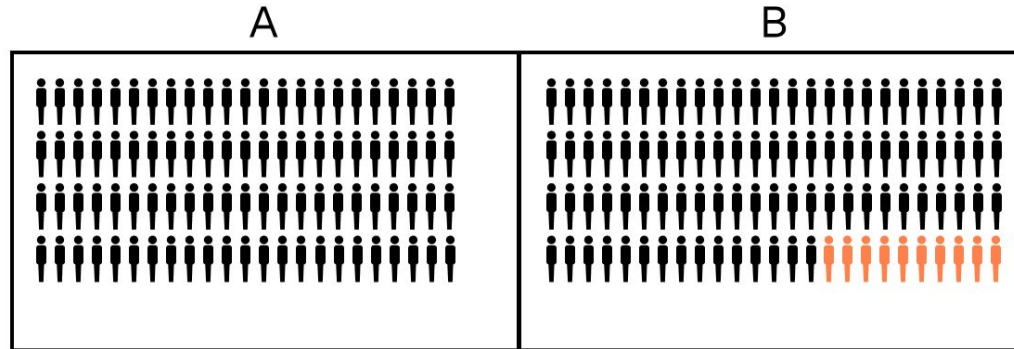


10 individuals are moved from district B to equalize the populations

PRISON GERRYMANDERING EXAMPLE^[1]



100 individuals can participate in elections in district A, compared to only 90 individuals in district B.



Vote strength

$$\frac{10 \text{ residents}}{9 \text{ voters}} = 1.1$$

PRISON GERRYMANDERING REFORM

In 2016, a large new prison was slated to open in Trousdale County, where the projected prison population would have accounted for 3 whole districts on its own.^[3]

A new state law was passed:

“When a reapportionment is made, residents of a correctional institution who cannot by law register in the county as voters may be excluded from any consideration of representation.”^[4] (TN Code § 5-1-111)

PRISON GERRYMANDERING REFORM

14 states

have taken action to avoid prison gerrymandering in
state-level redistricting.

Tennessee is not among them.^[5]

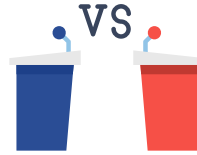


RESEARCH QUESTIONS



VOTE STRENGTH

Does the presence of one or more prisons significantly increase the "voting strength" for voters in that district?



DEMOGRAPHICS

Do districts with prisons differ from districts without prisons?



REAPPORTIONMENT

Are districts still substantially equal after reallocating the prison population?

DATA

- State legislative district boundaries came from [Redistricting Data Hub](#).
- Population data for TN state legislative districts are drawn from 2023 CVAP (Citizen Voting Age Population) datasets from [Redistricting Data Hub](#).
- Demographic info for TN state legislative districts comes from voter registration data gathered during the 2024 election (also from [Redistricting Data Hub](#)). This data was specific to the precinct level; it was aggregated to state legislative district using precinct mapping data from [Dave's Redistricting](#).
- Partisan data was drawn from [Dave's Redistricting](#) and is a composite of the following races:
 - President 2020
 - President 2024
 - Senator 2020
 - Senator 2024
 - Governor 2022
- TN state prison data (address, capacity, and year opened) comes from [TN.gov](#). Any information not available on [TN.gov](#) was obtained from [Wikipedia](#).
 - Prisons were geolocated using the R tidygeocoder library



DATA EXPLORATION

[R Shiny App](#)

VOTE STRENGTH

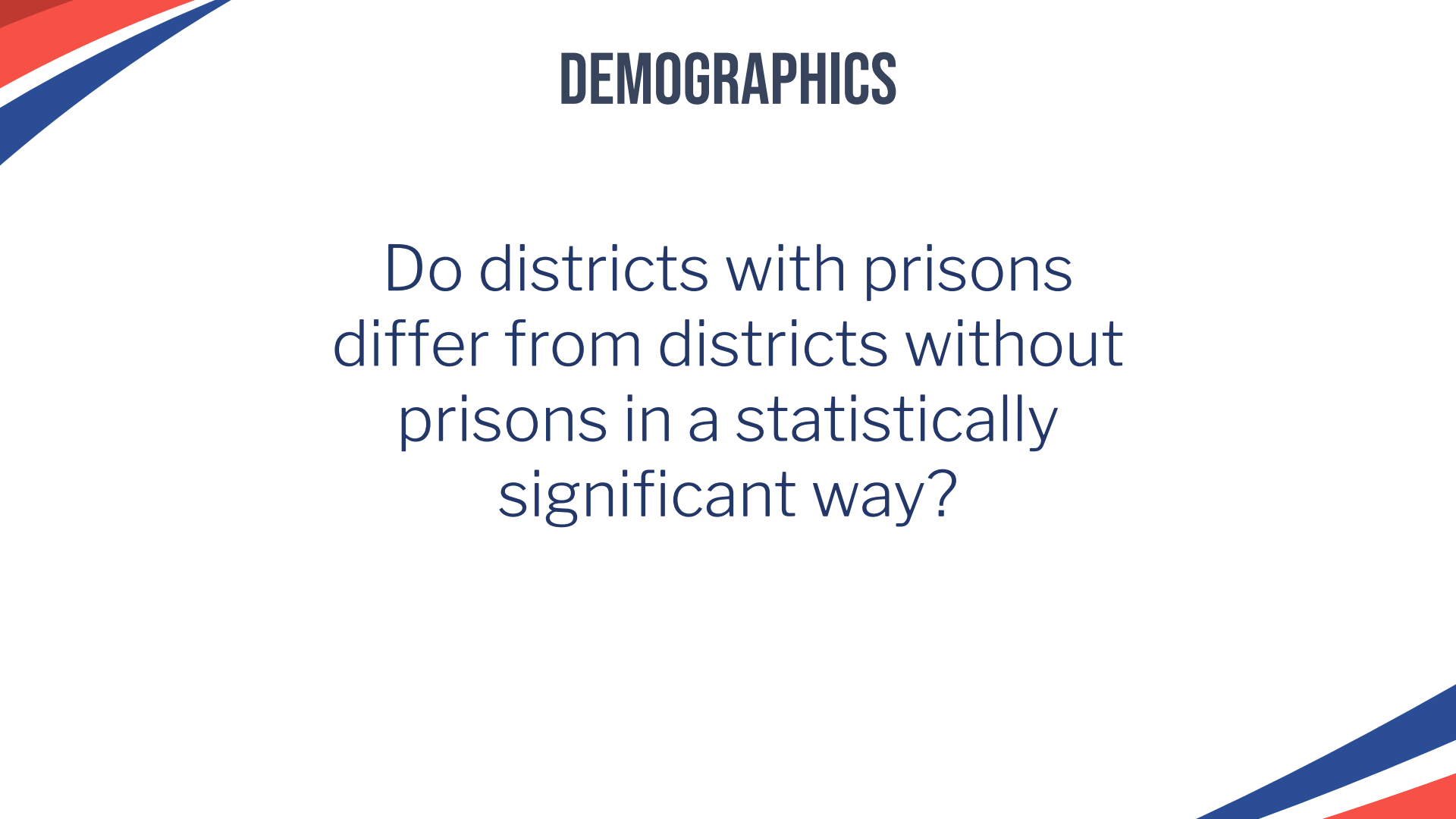
Do prisons significantly
increase voting power in the
districts where they are
located?

VOTE STRENGTH

For each additional prison, voting strength increases by an average of **0.03** for state house districts and **0.01** for state senate districts.

These results are highly statistically significant.





DEMOGRAPHICS

Do districts with prisons
differ from districts without
prisons in a statistically
significant way?

DEMOGRAPHICS

RACIAL MAKEUP

Percent white



GENDER MAKEUP

Percent male



PARTISAN LEAN

Percent Democratic

AGE MAKEUP

Percent under 30



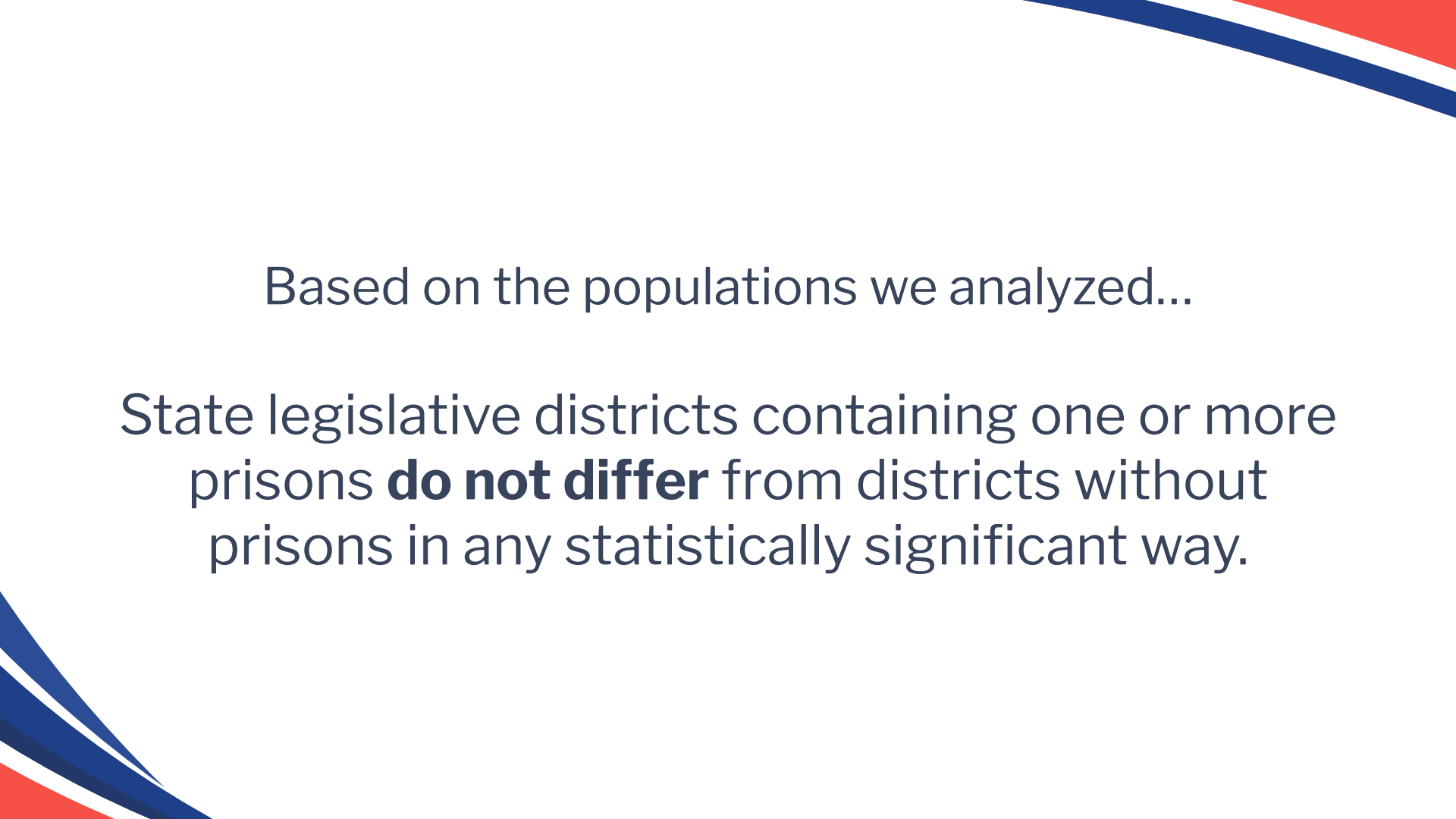
AGE MAKEUP

Percent over 65



POLITICAL ENGAGEMENT

Voter turnout in
2024 election



Based on the populations we analyzed...

State legislative districts containing one or more prisons **do not differ** from districts without prisons in any statistically significant way.



REAPPORTIONMENT

What happens if we redistribute
Tennessee's prison population
among all districts?

REAPPORTIONMENT

“The apportionment of senators and representatives shall be **substantially according to population.**”^[6]

-TN State Constitution, Article II Section 4

“[One can reasonably surmise that a **line has been drawn at 10%**—deviations in excess of that amount are apparently acceptable only on a showing of justification by the State; deviations less than that amount require no justification whatsoever.”^[7]

-White v. Regester, 412 US 755 - Supreme Court 1973

REAPPORTIONMENT

State Senate

District	Population Deviation	Population Deviation (Adjusted)
30	-8.42%	-8.15%
22	5.39%	5.7%
17	4.71%	5.02%
27	4.66%	4.97%
33	-4.86%	-4.57%

State House

District	Population Deviation	Population Deviation (Adjusted)
80	-5.96%	-10.67%
68	9.77%	10.1%
67	9.24%	9.57%
46	9%	9.33%
77	-6.75%	-8.99%

REAPPORTIONMENT

After reapportionment, do districts with high population deviation have any demographic trends?

REAPPORTIONMENT

State Senate

For each additional percent of population deviation....

- The percentage of the population that is under 30 **increases** by an average of 0.136%.
- The percent of the population that is white **increases** by an average of 2.7%.
- Voter turnout **decreases** by an average of 1%.

State House

For each additional percent of population deviation....

- The percent of the population that is 65 or older **decreases** by about 0.36%.
- The percentage of the population that is male **increases** by 0.32%, on average.

CONCLUSIONS

VOTE STRENGTH

Vote strength increases for districts with prisons, but the **effect is small**.

DEMOGRAPHICS

Districts with prisons **do not** differ from districts without prisons in a statistically significant way.

REAPPORTIONMENT

After reallocating the prison population, 2 districts **exceed a 10% deviation** from the target population. Districts with high population deviation have **some demographic trends**, but the effect is small.

THANKS!

Any questions?



Click here for
GitHub repo

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WORKS CITED

- [1] “Prison Gerrymandering”, *Redistricting Data Hub*, 2026, <https://redistrictingdatahub.org/resources/prison-gerrymandering/>
- [2] “Tennessee Profile”, *Prison Policy Initiative*, 2024, <https://www.prisonpolicy.org/profiles/TN.html>
- [3] “For the first time ever, Tennessee counties can avoid prison gerrymandering”, *Prison Policy Initiative*, 2021, https://www.prisonersofthecensus.org/news/2021/11/04/tn_redistricting/
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- [6] *Tennessee, Congress. CONSTITUTION OF THE STATE OF TENNESSEE*, Article II, Section 4, <https://redistricting.lls.edu/wp-content/uploads/TN-constitution.pdf>
- [7] *White v. Regester*, 412 US 755 - Supreme Court 1973